Design and Manufacturing of Ultra-Thin Thermoplastic Composites for Coilable Longerons

Federico Benazzo* and Sergio Pellegrino†
Graduate Aerospace Laboratories, California Institute of Technology, Pasadena, CA, 91125

Thermoplastic composites (TPCs) offer many advantages compared to thermosets in terms of mechanical properties, reprocessability, and flexible manufacturing. However, their implementation in aerospace systems has lagged behind due to manufacturing challenges, insufficient availability of ultra-thin prepreg materials, and lack of information on material processing. The current study aims to fill this knowledge gap by demonstrating a design strategy for manufacturing and testing TPC Triangular Rollable and Collapsible (TRAC) longerons. To explore the design space, both analytical and numerical models of the longeron's coiling were implemented, using experimentally determined mechanical properties of an ultra-thin (55 μm lamina thickness) PEEK prepreg. The results indicated that a [0/90/0] laminate could only be coiled to a 50 mm radius without failure due to the formation of a buckle, which acted as a stress concentrator. To verify these results, a TRAC longeron was then manufactured by exploiting the re-shaping capabilities of the matrix after initial consolidation, while the shape accuracy of the longeron and bond quality within the web region of the TRAC were assessed to identify any imperfections. The TRAC laminate was then coiled under the same conditions of the numerical model and the buckle formation was recorded and post-processed using image correlation software to compare directly with the simulation.

I. Introduction

THERMOPLASTIC composites (TPCs) have drawn increasing attention from the aerospace industry for several advantages over the more common thermoset-based materials. For instance, the reprocessability and re-shaping capabilities of TPCs provide an avenue towards a reduced environmental impact of composite structures. In addition, many TPCs offer improved mechanical properties, namely stiffness, strength [1], and fracture toughness [2], which greatly increase the loads sustained by structures before crack formation and enhance the resistance to crack propagation. Most importantly, thermoplastic polymers can be rapidly melted and bonded, providing a number of feasible methods for out-of-autoclave (OOA) manufacturing, which may greatly reduce costs, manufacturing times, and challenges with scaling the component's size. TPCs are particularly attractive for spacecraft structures for two reasons. First, there has been a growing interest in In Space Assembly and Manufacturing (ISAM) [3] and TPCs have been the focus of studies for their welding capability [4], which could be expanded to the in-space manufacturing of such ultra-thin composites. Second, large coilable composite structures are required to increase the spacecraft's surface area while adhering to the strict volume constraints of launch vehicles. One example is the Space Solar Power Project at Caltech [5], which envisages 85 m long deployable booms to unfold flexible photovoltaic tiles with a total area greater than 3000 m^2 for a single spacecraft, which is far greater than the power generation of any currently produced spacecraft. The basic structural element for this structure's deployment is the Triangular Rollable and Collapsible (TRAC) longeron, defined by the cross-section shown in Figure 1. To achieve a high packaging efficiency and also maintain bending stiffness, the flattening and folding curvatures of the longeron($1/r$ and $1/R$, respectively) are on the order of 50-100 m^{-1} in the two principal directions, which, due to the limited yield strain of composite materials, pose a strict limit on the thickness of the laminate to avoid material cracking during the folding process.

A previous study [6] has demonstrated that TRAC longerons can be manufactured from small thickness laminas of aerospace grade TPCs, namely polyetheretherketone (PEEK) reinforced with carbon fiber. PEEK is a prime choice as it displays mechanical properties similar, if not superior to comparable thermoset polymers, and has been extensively

*Graduate student, Graduate Aerospace Laboratories, 1200 E. California Blvd., Mail Code 105-50. fbenazzo@caltech.edu

†Joyce and Kent Kresa Professor of Aerospace and Civil Engineering, Jet Propulsion Laboratory Senior Research Scientist; Co-Director, Space-Based Solar Power Project, Graduate Aerospace Laboratories, 1200 E. California Blvd., Mail Code 105-50. AIAA Fellow. sergiop@caltech.edu

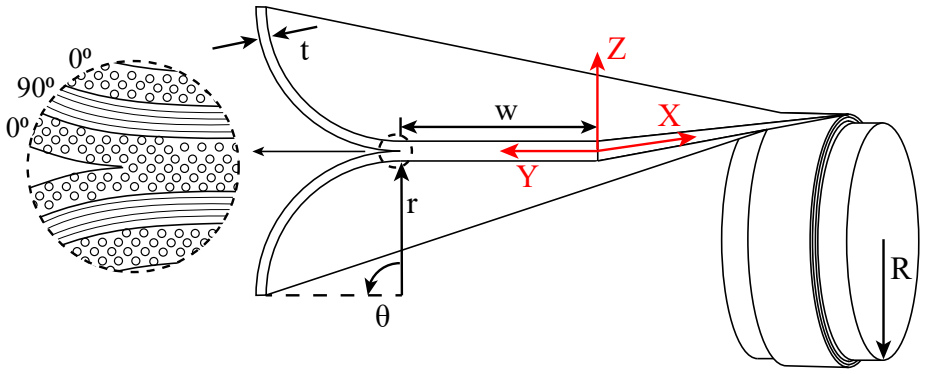

Fig. 1 TRAC cross-section geometry and coiling.

studied in the literature. However, manufacturing structures based on this polymer is challenging due to the high processing temperatures (~400°C) and viscosity, which amplify the effects of thermally induced residual stresses [7] and increase the complexity of fiber impregnation. Higher temperatures and viscosity are significant hurdles for ultra-thin prepreg manufacturing, which currently limit the availability of pre-impregnated carbon fiber tapes lighter than 50 gsm. For these reasons, the longerons produced in the previously mentioned investigation were prepared by dry fiber stacking of individual carbon fiber and PEEK layers, which is considerably more labor intensive than prepreg technologies and is thus unsuitable for large-scale manufacturing. The same authors also leveraged the asymmetry in the laminate to produce curved flanges through residual thermal stresses, which restricts the range of curvatures of the flange that can be achieved without changing the composite's volume fraction or thickness.

The mechanical performance of semi-crystalline thermoplastic polymers, including PEEK, also shows a significant dependence on their degree of crystallinity (DoC), which has been shown to influence the stiffness and strength of TPCs [1] due to the varying ratio between the brittle crystalline phase and the ductile amorphous phase. Less focus has been placed however on how the DoC varies the processability of the composite material due to the thermal hysteresis of the matrix' morphology. For instance, to achieve sufficient mobility of the tightly packed and organized polymeric chains of the crystal phase the material must be processed at temperatures above the melting point, while the amorphous phase, with its looser intermolecular bonding, can be processed at lower temperatures [8]. Finally, the difference in coefficient of thermal expansion between the amorphous and crystalline phases is significant enough to vary the magnitude of residual thermal stresses present in the material after consolidation during the cooling process [7]. Thus, maintaining a low DoC throughout manufacturing may allow processing at lower temperatures and shorter times, thus leading to simpler processes.

The present investigation addresses the challenges and potential of TPCs by designing a PEEK-based laminate, with currently available prepregs, that does not yield under the curvature changes required for coiling. The design process is based on the mechanical properties of a PEEK-carbon fiber prepreg, presented in Section II.B, and implements them in both a linear analytical model and a detailed coiling analysis using Abaqus finite-element formulation, which are both described in detail in Sections II.C-II.D. The same PEEK-carbon fiber prepreg is then used in a novel fabrication method, presented in Section III.A, that leverages the benefits of thermoplastic polymer chemistry to produce TRAC longerons. The shape and microstructure of the coilable structure are then evaluated via laser scanning and computed microtomography (μ-CT), respectively; the results of the two analyses are discussed in Sections III.C and III.D. Finally, the manufactured longerons are coiled to validate the modeling scheme in Section IV. Section V concludes the paper.

II. Design of Thermoplastic Laminate

A. Materials and Manufacturing

The thickness of a deployable thin shell controls the maximum stress in the shell when coiled. Thus, the areal weight of the carbon fibers and PEEK matrix system - which generally correlates to the final lamina thickness - was used as the criterion for selecting the optimal prepreg to manufacture the composite laminate. One of the lightest available prepregs, supplied by Sakai Ovex Ltd., was used for the present investigation. The prepreg is sold as a 60 mm wide tape with a total areal weight of 68 gsm (38 gsm for CF and 30 gsm for PEEK), and it is manufactured by spread-towing from 18K tows of M30SC carbon fibers. Additionally, a 10 μm thick film of semi-crystalline PEEK was purchased from BARplast LLC to supplement the resin content of the prepreg during the manufacturing of TRAC longerons, as explained in Section III.A.

Flat UD 4-ply laminates of this prepreg were compacted in an autoclave, using Kapton vacuum bags. The processing cycle included a ramp to 365°C, followed by a dwell of 1 hour at 5.5 bars, and ended with a cooling ramp at 2°C/min. Coupons for mechanical testing were cut using a sharp blade and a ZEISS Xradia 510 Versa 3D μ-CT scanner was used to image the samples and measure the thickness, volume fraction, and void concentration. A sample image of the thickness cross-section, presented in Figure 2, shows a consistent thickness and variable fiber-matrix distribution. This is due to the high viscosity of the matrix, which under the relatively low autoclave pressure when compared to hot-press manufacturing, does not allow for sufficiently uniform dispersion of the fibers. The averaged values of thickness and volume fraction are listed in Table 1.

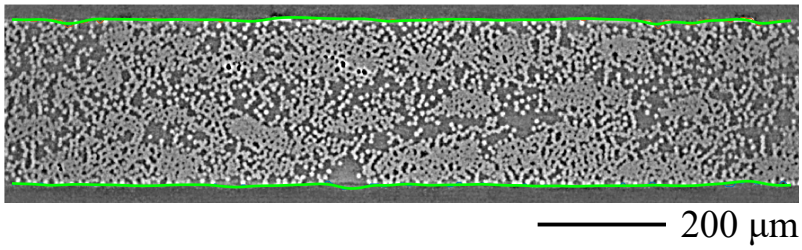

Fig. 2 Gray-scale image of μ-CT scan of UD laminate at 1 μm per pixel resolution with green lines showing the identified top and bottom boundaries of laminate.

B. Mechanical Characterization of UD Laminates

To create numerical and analytical models for the design of the composite laminate, the relevant mechanical properties had to be evaluated. First, tensile tests of both UD and ±45° samples were performed to measure the Young's modulus and, the shear strength. Due to the limited width of the prepreg tapes and challenges with bonding adjacent tapes, the coupons' size was limited to 120 × 15 mm and 60 × 15 mm for the 0° and 90° oriented specimens, respectively, while the shear specimens were manufactured to 250 × 25 mm to reduce the effects of the clamped edges. The specimens were bonded to 280 grit sandpaper tabs, which were sized based on length ratios from ASTM D3039 [9], while a central 30 mm long region of the samples was speckled with matt-white paint for Digital Image Correlation (DIC) measurements, performed with stereo-mounted digital cameras. The load was applied by an Instron electromechanical testing machine at a rate of 0.1 mm/s and 0.05 mm/s for the 0°/±45° and 90° specimens, respectively. For the off-axis tensile tests, the shear modulus was obtained from the initial linear response while the strength was defined by the change in slope of the curve [10], as this indicates the yielding of the material and successive load increments correspond to the plastic deformation of the cross-ply laminas. The results of the longitudinal tensile tests to determine the material properties along the fibers are presented in Figure 3, where the solid lines corresponds to the mean values calculated from the tests, the dashed lines represent the fitted linear curves, and the shaded areas correspond to the standard deviation amplitude.

For the calculation of the longitudinal and transverse strengths of the composite, a test with loading conditions similar to those experienced during the packaging of the longerons was selected. Thus, 60 × 20 mm coupons were cut and speckled for a parallel platen test (the apparatus and test method are described in Yee and Pellegrino [11]) in which

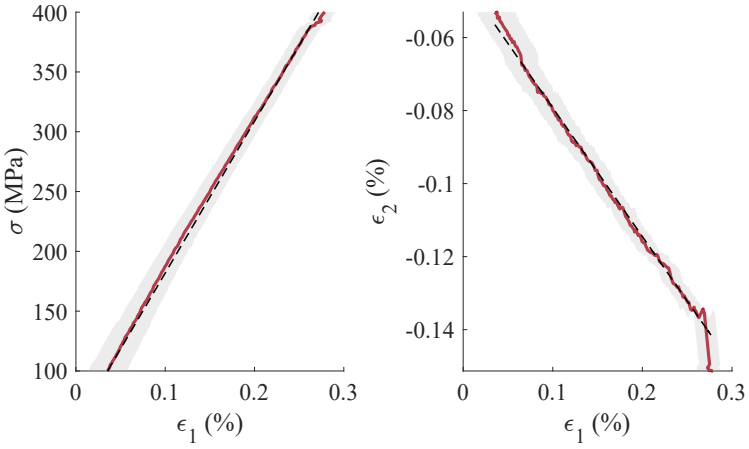

Fig. 3 Raw data from tensile tests on UD composite to obtain stiffness modulus along fiber direction and Poisson ratio.

curvature measurements were carried out on both the compressive and tensile sides with two sets of synchronized digital cameras placed similarly to Schlothauer et al. [12, 13]. The curvatures measured at yield on both compression and tension sides were substituted in equations 1-2, along with the previously measured properties to obtain the bending strengths.

$$X_{T,C} = \frac{E_1 t}{2(1 - \nu_{12}\nu_{21})} K_{1\,T,C} \tag{1}$$

$$Y_{T,C} = \frac{E_2 t}{2(1 - \nu_{12}\nu_{21})} K_{2\,T,C} \tag{2}$$

In these formulations, the indices 1 and 2 indicate the local coordinate system, aligned in the direction of the fibers, while the subscripts C and T correspond to the compression and tension sides of the sample. In Figure 4, two representative snapshots of a longitudinal sample show the curvature distribution measured by the DIC apparatus. The highest curvature measured in the central section of the specimen was selected as the failure curvature under tension in each test. The yield strengths and stiffness moduli obtained from tests on six samples are listed in Table 1, where S_{12} represents the shear strength.

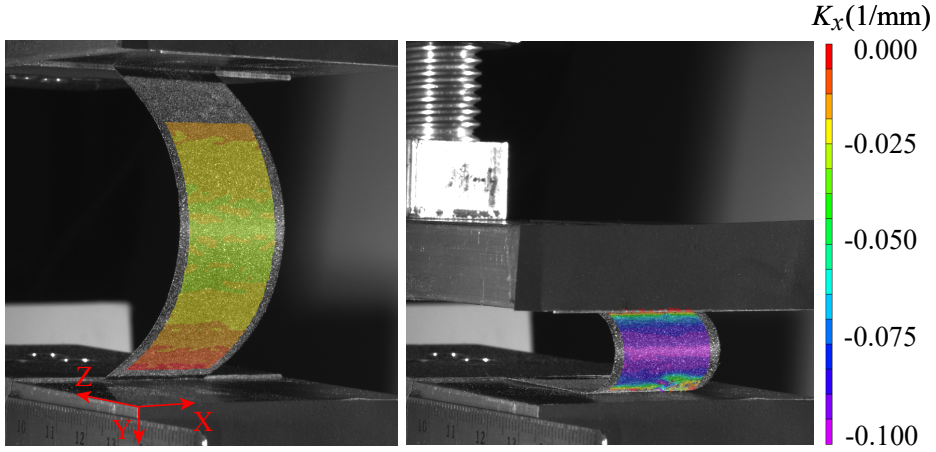

Fig. 4 Snapshots of sample during bending test with overlaid curvature measurements from DIC.

Fibers	M30SC
Matrix	PEEK
Prepreg nominal thickness	55 μm
V_f	0.49
E_1	127 GPa
E_2	5.8 GPa
G_{12}	5.2 GPa
ν_{12}	0.35
X_T	2,819 MPa
X_C	2,805 MPa
Y_T	69 MPa
Y_C	67 MPa
S_{12}	49 MPa

Table 1 Material properties of four-ply PEEK-carbon fiber laminate.

C. Laminate Design via Analytical Model

Deployable longerons should be able to perform two tasks. First, flatten and coil with the material remaining in the elastic regime and, second, provide the maximum longitudinal bending stiffness in the deployed configuration. Thus, an analytical model based on Classical Lamination Theory (CLT) was developed to evaluate the stress distribution during packaging, based on the lamina's mechanical properties, layers' orientation angles, and thicknesses.

The cross-section is subjected to different loading conditions in the flanges and the web. The flange undergoes simultaneous curvature change about the y-axis for flattening and the x-axis for coiling (Figure 1), while the web region is only coiled about the x-axis. Hence, the stress distribution for both regions was calculated and the highest stresses were then compared to the measured yield strengths with the Hashin 2D composite failure criterion [14] to identify failure through the layers' thickness 3:

$$
\begin{aligned}
F_{ft} &= \left(\frac{\sigma_{11}}{X_T}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 \geq 1 \qquad & F_{mt} &= \left(\frac{\sigma_{22}}{Y_T}\right)^2 + \left(\frac{\sigma_{12}}{S_{12}}\right)^2 \geq 1 \\
F_{fc} &= \left(\frac{\sigma_{11}}{X_C}\right)^2 \geq 1 \qquad & F_{mc} &= \frac{1}{4}\left(\frac{\sigma_{22}}{S_{23}}\right)^2 + \frac{Y_c^2\sigma_{22}}{4S_{12}^2 Y_c} - \frac{\sigma_{22}}{S_{12}} - \left(\frac{\sigma_{12}}{S_{12}}\right)^2 \geq 1 \\
F_{Hashin} &= max(F_{ft}, F_{fc}, F_{mt}, F_{mc})
\end{aligned}
\tag{3}
$$

This approach was implemented in an algorithm that varied the lamina thickness and coiling radius to determine which combination of the two would not incur failure in a cross-ply [0/90/0] laminate configuration, which exhibits high longitudinal stiffness thanks to the two 0° layers and sufficient transverse stiffness for deployment. Of the remaining design parameters, w and θ were chosen as the previously reported values of 8 mm and 90° [15], respectively, as these do not influence the applied curvatures during coiling. The flange radius r does influence the flattening curvature, but it was also kept constant at 13.6 mm to leave the stiffness and buckling limit of the longeron constant.

The results of the parametric study are shown below in Figure 6. They indicate that the failure of the laminate is bi-modal, i.e. for coiling radii small enough the material in the web (which is twice as thick as the flange), reaches the maximum allowable stress due to the increasing curvature about the x-axis. Instead, for larger coiling radii the flanges dominate the failure as the curvature about x becomes less significant. Thus, for any chosen lamina thickness, there is a point where the magnitude of F_{Hashin} is minimized and, for sufficiently thin prepreg laminas (between 40 to 100 μm) this minimum is less than 1, indicating no failure of the material. Since the selected PEEK prepreg has a measured compacted thickness of 55 μm, the minimizing coiling radius is 28 mm, for which the F_{Hashin} value is 0.45.

D. Numerical Model for Validation

The analytical model presented in the previous section is a powerful yet simple tool to perform parametric studies to identify the longeron geometric configuration and laminate designs that satisfy the requirements. However, the model considers the flange and web regions separately, neglecting the interaction between the two. It has been shown [16] that during coiling a buckle forms near the transition between the coiling hub and the deployed longeron, forming a region of increased local curvatures. The shape and size of this buckle are dependent on the material parameters, cross-sectional geometry of the longeron, and coiling radius. Thus, a more detailed numerical model is necessary to identify the stress concentration at the buckle to avoid premature failure of the material.

A numerical simulation was implemented in Abaqus 2024, modeling the coiling of a TRAC longeron on a rigid hub of radius R. The longeron is modeled with S4R elements with the geometry defined in Section II.C, while its length is a function of R to minimize the number of elements in the simulations with small coiling radii. The hub is translated towards the longeron to bring them into contact, while a pressure normal to the top flange is applied to it. A longitudinal tension is also applied at the tip of the longeron (Figure 5), while the hub is rotated counter-clockwise to coil the flattened longeron. It can be noted in the numerical results presented on the right of Figure 5 that there is a region of uniform K_x, which corresponds to the fully coiled longeron around the rigid hub of radius R. This uniform region extends into the deployed longeron through a complex transition region that includes the aforementioned buckle. The large out-of-plane displacement of the buckled flange causes a local increment in curvature in both principal directions, as well as twist.

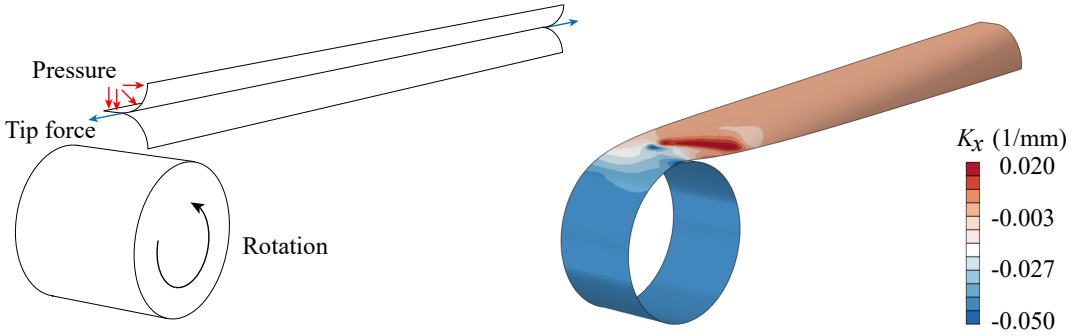

Fig. 5 Simplified diagram of numerical model set-up alongside the curvature change on the bottom flange during coiling.

To evaluate the failure margin of the material during coiling considering this increased curvature, the through-thickness stress distribution in the buckled region was calculated, and the same Hashin criterion defined in Eq. 3 was applied. The results for several lamina thicknesses and coiling radii are presented in Figure 7, however, the whole design space investigated in the analytical model could not be reproduced due to convergence issues at low radii and thick laminas. The measured failure values in the numerical simulation are between 2 and 5 times higher than predicted by the analytical model, with a non-linear increase as one approaches the bottom right corner of the plot in Figure 7. This is because of the cubic relationship between thickness and bending stresses when moving along the x-axis of the plots, and because of the non-linear relationship between the coiling radius and the shape of the buckle when moving along the y-axis.

It is clear from this comparison that the simple analytical model cannot correctly predict the coiling failure, and is not a suitable tool for reliable design when considering failure. For the current application, the numerical model predicts that for a TPC prepreg with 55 μm lamina thickness, the material remains in the elastic regime if coiled to a radius of 50 mm or greater. This indicates that if the appropriate packaging parameters are chosen, currently available PEEK-based carbon fiber prepregs are suitable for use in a compact coiling apparatus.

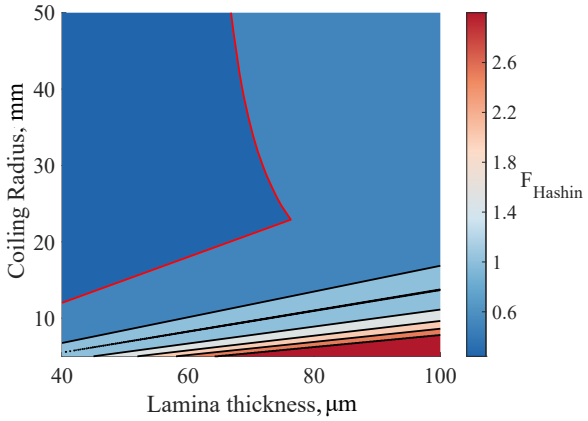

Fig. 6 F_{Hashin} **in analytical model.**

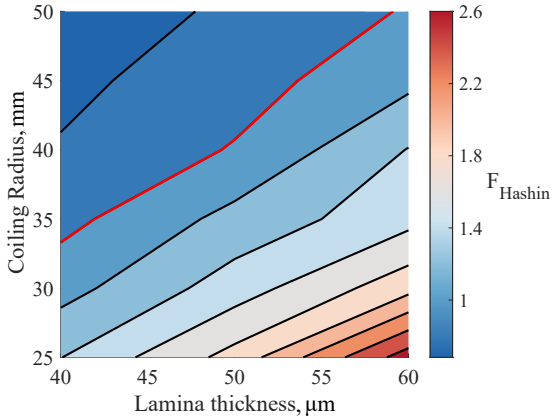

Fig. 7 F_{Hashin} **in numerical model.**

E. Coiling numerical analysis of extended length longerons

The coiling simulations have been so far employed to determine the minimum coiling radius for a certain thickness laminate. However, considering the objective of developing OOA manufacturing techniques for long deployable longeron composites, other inherent questions may be answered with the numerical model. Thus far coiling simulations in this and other investigations [16–18], have only observed the buckle shape and influence up to 1.5 revolutions and for a longeron of length 1 m. Yet, when considering the large deployable structures referenced in the Section I, it is clear that an improved understanding of how the buckle develops with number of hub rotations is crucial. Thus, a 5 meter long TRAC longeron, with the same material description, geometry, and layup and previously reported in Section II.C was simulated. The selected hub radius and lamina thickness were 50 mm and 50 μm, respectively, as these proved to be below the failure criteria for the laminate and also improved convergence of the simulation.

The simulated longeron was coiled for a total of 4 times around the central hub, and the shape of the buckled region was monitored through the process with representative snapshots shown in Figure 8. The contours show that the buckles is fully developed when the first revolution is completed, expanding in size and amplitude of curvature from the initial buckle formation. However, the shape remains mostly invariant throughout the successive coiling rotations. This buckled shape is characterized by a large concavity, corresponding to the deep blue region (location 1), which is adjacent to regions of localized positive curvature where the flange meets the hub, corresponding to the deep red regions (location 2). These peaks in curvature, both the concave and convex ones, appear to be increasing in amplitude with additional rotations, but only marginally compared to the difference between the nominal curvatures due to flattening and coiling.

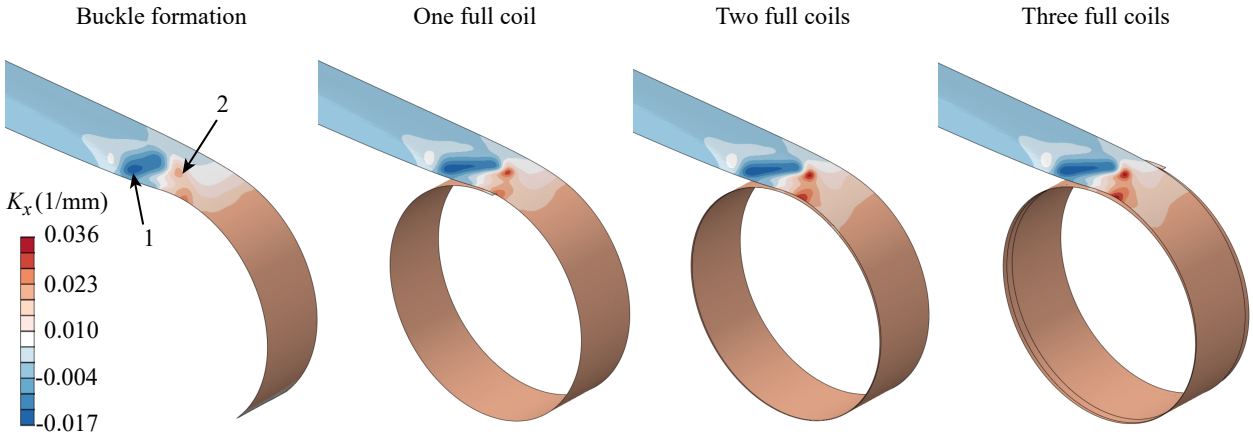

Fig. 8 Evolution of buckle with successive coiling rotations for 5 m long longeron.

Focusing on the evolution of the curvature along the length of the longeron during the repeated rotations, displayed in Figure 9, one can notice the same trend presented above. The greatest change in curvature takes place within the first revolution, and what appears to be half-way through the second. Nonetheless, both curvatures and corresponding stress in the material continue to grow throughout the coiling experiment. Between 10 and 25 radians, K_x grows in magnitude by 1.5% and 3.5% for location 1 and 2, respectively. This is a small increment, but its significance is based on the notation that this curvature is expected to decrease due to each successive layer of material increasing the radius of curvature of the hub by approximately 150 μm with each rotation. Instead, the buckle is marginally increasing in its depth and the resulting curvature gradient through the composite flange as it is coiled more. This indicates that assuming a constant stress due to the buckled region as the deployable structured is scaled may lead to underestimate the stresses in the material.

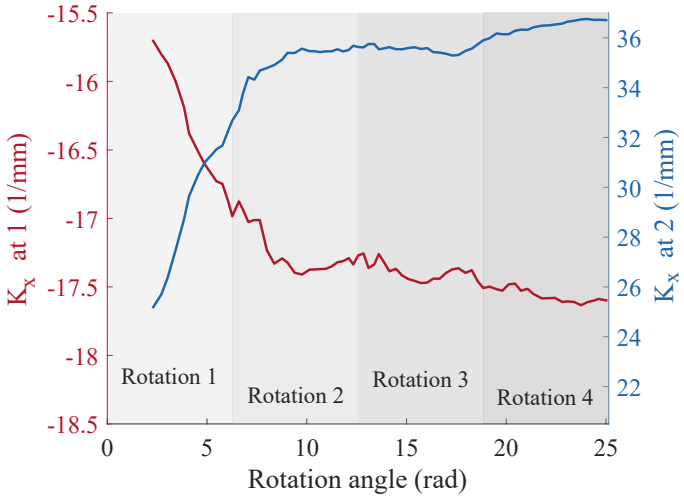

Fig. 9 Evolution of curvature at two highest stress locations in the buckled region with successive coiling rotations.

III. Manufacturing and Analysis of Thermoplastic Longerons

A. Leveraging the Degree of Crystallinity

TPCs offer considerably more manufacturing flexibility over thermoset composites, mainly thanks to their ability to be reprocessed after the first fabrication cycle. This is true for all thermoplastics when processed above the melting temperature, but it is particularly interesting in semi-crystalline thermoplastics, including PEEK, which exhibit a crystalline phase that when formed during thermal cycling remains stable and unvaried after post-processing, and an amorphous phase which, due to its less organized molecular arrangement and weaker bonding interactions, is mobile at lower temperatures, allowing for reshaping and new crystal formation. This characteristic was exploited in the manufacturing process by subjecting the compacted laminate to a second thermal cycle in which the laminate was shaped into the TRAC cross-section, to freeze the remaining amorphous phase of the thermoplastic polymer into the new shape. These new fabrication steps eliminate the need for a second autoclave cure to bond together the two parts of the TRAC geometry [19], thus reducing the quantity of consumables and avoiding the creation of a discontinuous bond that is likely to act as a weak point for the propagation of delamination cracks.

This novel fabrication technique involves two thermal cycles, one requiring pressure in the autoclave and one in the oven, and is summarized in Figure 10. The first step is a compaction of the six composite layers (the aforementioned $[0/90/0]_s$) which need to be split into two 3-layer laminates to form the flange. To achieve this, a 25 μm Kapton layer treated with release agent was placed at the laminate's mid-plane and extending to three quarters of its width. To compensate for the increased thickness, the 10 μm thick film of semi-crystalline PEEK was also placed at the midline in the opposite quarter. The laminate was sandwiched between a 25 mm thick aluminum plate and a 700 μm thick steel caul plate cut to the exact laminate dimensions. Both metal surfaces were covered with Kapton films

covered in release agent to improve the surface finish and prevent sticking. Finally, the layup was enclosed in a vacuum bag and processed as described in Section II.A. After the thermal cycle, the Kapton layer was removed and the non-bonded layers were bent outward into the TRAC cross-section and made to adhere to aluminum molds. Finally, the laminate was restrained using thin steel sheets to conform to the molds and set in an oven at 370° C for 20 min. The resulting longeron, shown in Figure 11, was trimmed to obtain the specified subtended angle θ and web width.

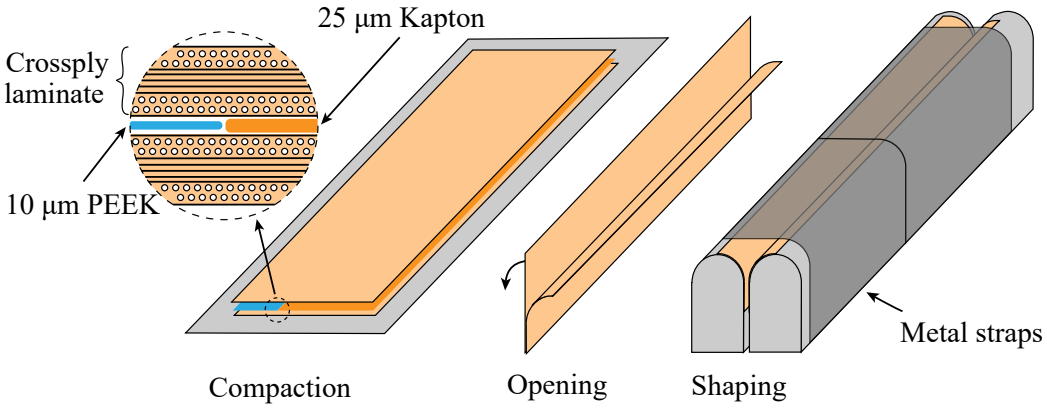

Fig. 10 Manufacturing steps of TPC TRAC boom.

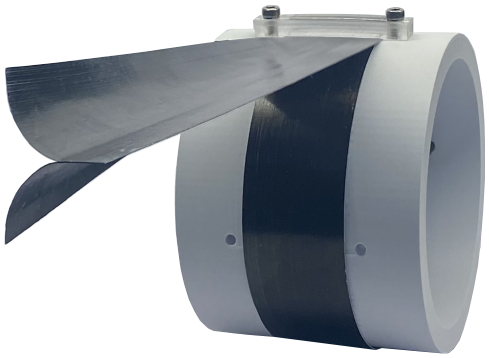

Fig. 11 Manufactured TRAC longeron, coiled on a mandrel.

B. Relationship Between Processing Parameters and Degree of Crystallinity

As stated in the introduction, the DoC influences the processability of thermoplastic polymers. The present study introduces a novel manufacturing procedure to fabricate large aspect ratio deployable composite structures by leveraging the tunability of the crystallinity. In particular, it has been shown amorphous polymers can be processed at lower temperatures and with more ease than semi-crystalline thermoplastics [20], and thus maintaining a low DoC during the first fabrication steps is considered favorable to facilitate following shaping or welding processes. Therefore, a lower temperature cycle to 365°C (compared to the more common 395°C for PEEK processing) was implemented to reduce the crystallinity, and correspondingly increase the amount of amorphous phase. The amorphous phase within the polymer could then be reshaped with a subsequent heating cycle, that is still below the melting temperature, without affecting the already formed crystal phase. To confirm that these manufacturing steps allow for the reshaping of a laminate, the DoC was measured for laminates produced with the standard 395°C and the lower 365°C, as well as after the second process at 370°C. The cooling rate during processing was kept identical for all three laminates, at 2°C/min.

It has been shown [21, 22] that Fourier Transform Infrared (FTIR) spectroscopy may be used to measure the DoC of PEEK-carbon fiber composites by identifying the ratio between the peaks at 1305 cm^{-1} and 1280 cm^{-1}, which

correspond to the stretching of the carbonyl and diphenyl ether bonds, respectively. However, the correlation between this ratio and the DoC in these papers was empirically formulated with parameters that may not apply to all PEEK-carbon fiber composites. Yet, given the current objective of comparing the DoC between different samples with the same PEEK matrix, the aforementioned ratio of peaks can be used to infer which samples have the larger vs. the smaller crystallinities. Three measurements of the absorbance spectrum of each sample were performed with a Nicolet 6700 FTIR with diamond ATR attachment and averaged to provide the curves in Figure 12 and tabulated in Table 2. The ratios indicate that the DoC is indeed lower for the lower temperature cycle, and thus should have facilitated the OOA shaping of the laminate.

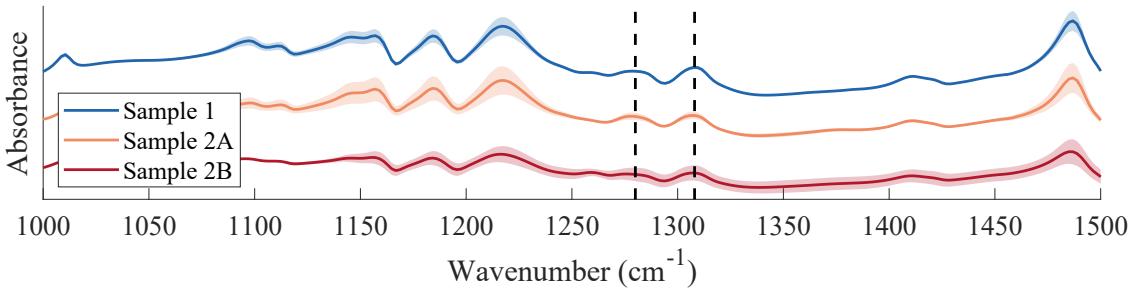

Fig. 12 Absorbance spectra for two samples overlaid with standard deviation shading.

Note that the ultimate objective of this research is not to employ low crystallinity PEEK-based composites for deployable structures. The proposed process with lower DoC is simply considered to ease the shaping and welding of the flanges after initial compaction of the laminas. Yet, to maintain the superior mechanical properties of high DoC PEEK, the longeron would undergo a crystallization step with slow cooling to achieve a higher level of crystallinity.

Sample #	Laminate process description	Mean	Std. Dev
1	Max temperature 395°C	1.115	0.011
2A	Max temperature 365°C	1.021	0.008
2B	Post-processed at 370°C	1.025	0.05

Table 2 Peak ratios indicating DoC for differently processed laminates.

C. Shape Analysis

To compare the above described manufacturing process with previously used methodologies for thermoset matrices [19], an in-depth analysis of the shape of the manufactured longerons was performed. First, the longeron was clamped at one end of the web and suspended from the edge of a table to avoid any deformations due to gravity. Both inner sides of the flanges were then imaged with a Faro Arm (Edge 14000) with a 3D laser scanner attachment, thus creating a point cloud of approximately 15 million points for the 540 mm long longeron. Cross-sections of the longeron at 100 mm intervals are shown in Figure 13, while a three-dimensional plot of the deviation of the manufactured structure from a perfect longeron is shown in Figure 14. The latter was generated by using the fmincon function in MATLAB to determine the best-fit cylinders to both of the flanges and the best-fit plane to the web. The distances of the scanned points from these fitted geometries were calculated as the error and averaged over a grid of 1×1 mm^2 squares to generate the surface variation shown in the 3D plot. The errors were divided by the nominal flange radius and web width (13.6 mm and 8 mm, respectively) to obtain relative error deviations.

The first plot shows that, despite the half-meter length of the longeron, there is no visible rotation between the end cross-sections, indicating that the twist along the y-axis is minimal. This is generally difficult to achieve with such thin and long structures with an open cross-section due to their small resistance to twisting moments, yet the current layup and material selection along with the novel manufacturing process proved successful in minimizing twisting. This is particularly crucial for the objective of scaling the longeron to tens of meters since twist compounds with length. Not to

mention the asymmetry that twist causes in the structure, leading to a reduction in the mechanical performance.

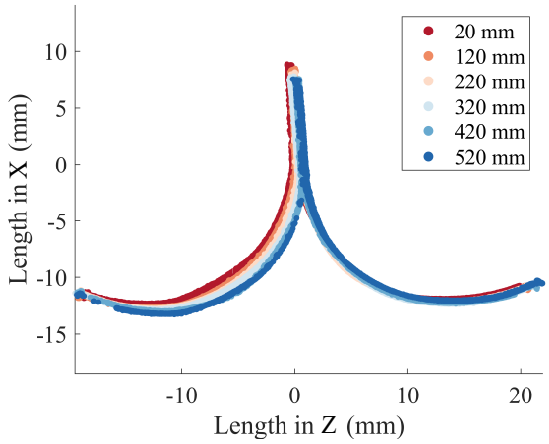

Fig. 13 Cross-section of TRAC from laser scanning at different y-intervals.

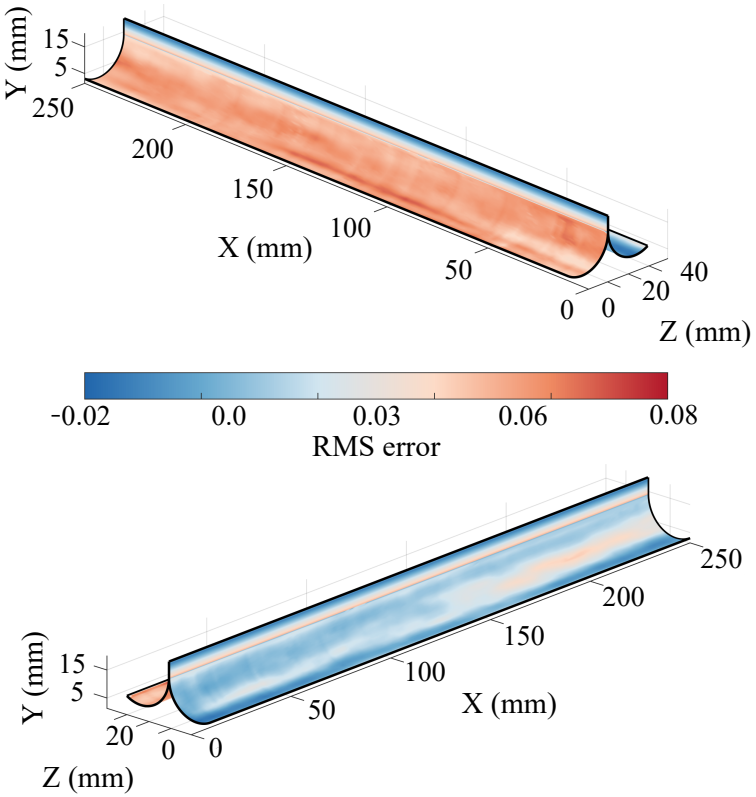

Fig. 14 Root mean squared error of the measured flange's radius and web inclination, compared to nominal values.

Similarly, the second plot shows that the deviations of the longeron from the desired geometry are small considering the length of the longeron. A maximum of 7% error is measured on the left flange, while the right flange displays a lower deviation from the best-fit cylinder, which reaches a maximum of 3%. The error being positive indicates that the the radius of curvature of the actual longeron is larger than that of the mold on top of which it was manufactured,

which means that the composite exhibits some degree of spring-back after the shaping process. This is in contrast to the previously reported spring-in deformation measured on TRAC longerons [18], which may be due to either the lack of autoclave pressure in the currently presented fabrication process, or due to residual thermal stresses, which have been shown to be more prominent in thermoplastic based composites [7]. Nonetheless, focusing on the magnitude of these spring-in and spring-back effects, the thermoplastic longerons show less than half the maximum deformation after fabrication compared to the thermoset ones, proving the efficacy of this new material and novel fabrication technique.

Considering the shape of the scanned longeron, it appears, based on the RMS error, that the left flange has a uniform radius of curvature, except for a 50 mm long section starting at 0 mm and located approximately at a subtended angle of 45 °, where the radius is smaller. Conversely, the right flange displays a reduction in its radius between 150 and 250 mm, yet still in the middle of the flange. Therefore, it is clear that the imperfections present in both flanges are longitudinal, i.e. they run along the X-axis rather than being localized at a specific X-value and spanning across the full range of subtended angles. This might indicate the presence of misalignment in either the placement of the 0° layers or during the shaping process when placed on the molds. Nonetheless, the generally low deviation from the perfect shape is a positive indicator that the TPC and manufacturing technique are suitable for creating ultra-thin deployable space structures at large scale with good shape accuracy.

D. Microstructure of Web/Flange Interface

Besides the shape accuracy of the fabricated longeron, the manufacturing process presented in this paper is also expected to improve the bond between the two halves of the structure. This conjecture stems from the fact that in the manufacturing of thermoset longerons [19] the two halves are bonded in a post-cure process with an additional glass-fiber plain weave lamina, while the current fabrication process involves the simultaneous compaction of all six laminas in the web in one thermal cycle. Therefore, the microstructure across the thickness of the web should not show any discontinuities, making it more resistant to delamination-type failure. μm-CT scans of the web-flange interface region were performed to image this region of the longeron. The tomographies were performed with the Y-axis of the longeron aligned with the axis of rotation of the CT scanner, meaning that exported cross-sections lie on the X-Z plane of the longeron.

First, a tomography with 2.5 μm per pixel resolution was performed and a representative cross-section is shown in Figure 15. This image clearly distinguishes between the two 3-ply flanges on the right-hand side and the 6-ply web on the left-hand side. The separation between the flanges is marked by the darker and increasingly wider region in the scan, while the variation in texture in the vertical direction depicts the changing orientation of the plies in the laminate. A second scan, with the resolution increased to 1 μm per pixel, shows the distinctions between the six plies more clearly, with the middle two plies combined with no discontinuity across the thickness as expected. There is a region of increased matrix content between the middle plies, possibly due to the added 10 μm PEEK film. This is desirable as the high fracture toughness of PEEK and the low content of fibers that may act as crack initiators should improve the mode-I fracture toughness along the composite midline.

Uniformity of the composite layup is crucial to ensure that when scaling of the longeron to tens of meters, there is no significant local defects and discontinuities. To evaluate the efficacy of the new manufacturing method in maintaining uniformity, the thickness of each layer and its variation along the laminate needed measuring. Thus a tomography scan of a 12.5 mm long section of the longeron was performed at the aforementioned resolution of 1 μm and the cross-sectional images were analyzed through an image processing algorithm in MATLAB. In the specific, the algorithm focused on one row of the array of pixels that composed the image, and fitted a normal distribution to the pixels' gray-scale values. Looking at a representative scan in Figure 16, it is clear that the 0° laminas, for which the scan shows the cross-sections of individual layers embedded in matrix, the pixels range from very bright in correspondence to the fibers and very dark in correspondence to the matrix between the fibers. Instead, the 90° layers are mostly gray due to the reduced signature the fibers have in the tomography when oriented normal to the main axis of the CT scan. Thus, when looking at the standard deviation of the gray-scale values in each row of the image, the spread is considerably larger for the 0° layers compared to the 90° as shown on the right of Figure 16. Thus, by selecting an appropriate cutoff value for the standard deviation σ, the image can be split in the 5 laminas.

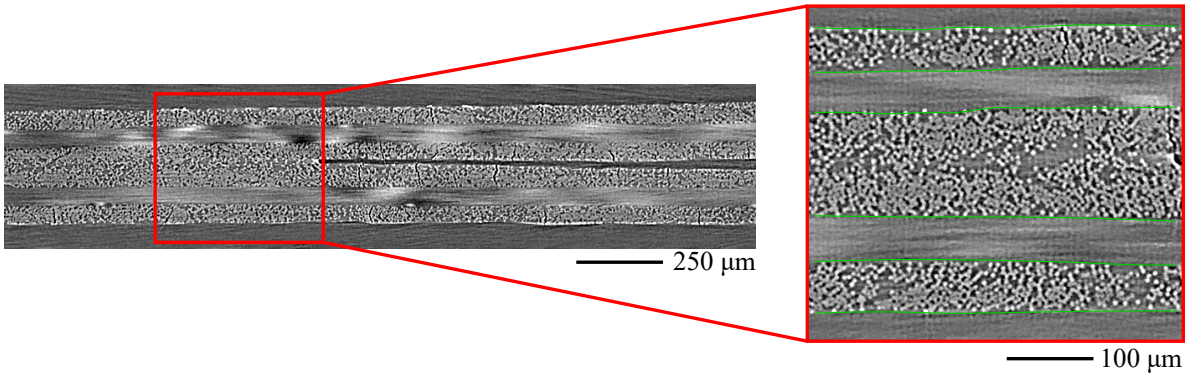

Fig. 15 μ**-CT scan of web-flange interface, with enlargement showing the layer separation in the web and with green lines representing boundaries between different orientation plies.**

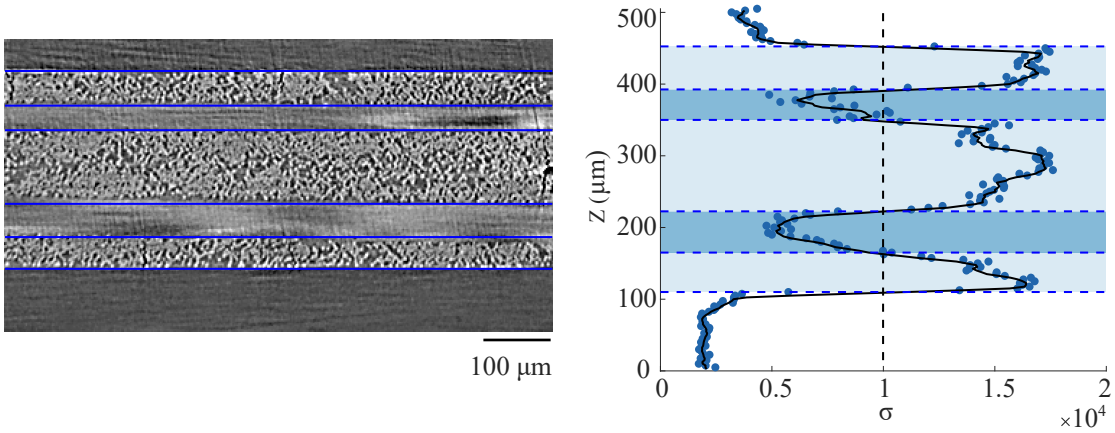

Fig. 16 On the left, μ**-CT scan of web region with overlaid lamina edges and, on the right, variation in standard deviation of pixel gray-scale value distribution for each row of the image.**

This image processing was applied to all 5000 cross-sections of the web region and the thickness variation of each layer is plotted in Figure 17. The thickness of each layer remains invariant in the 0° layers, with only noticeable change in the two 90°. These vary from approximately 55 to 40 μm in the thinnest regions. This is a considerable change, yet it occurs in the transverse plies, which have an insignificant effect to the bending stiffness of the material, thus indicating that such imperfections should not greatly affect the performance of the longeron. However, future development of the manufacturing method should address this thickness non-uniformity, perhaps by using higher precision molds and coul-plates. Finally, the average thicknesses of each lamina, starting from the bottom, are follows: 57, 56, 126, 45, 52 μm, with a total thickness averaging to 336 μm, which is less than 2% larger than the expected total thickness of a 6-ply laminate made with 55 μm thick prepreg, meaning that the numerical model's predicted failure radius for a 55 μm thick lamina is valid for the manufactured longeron.

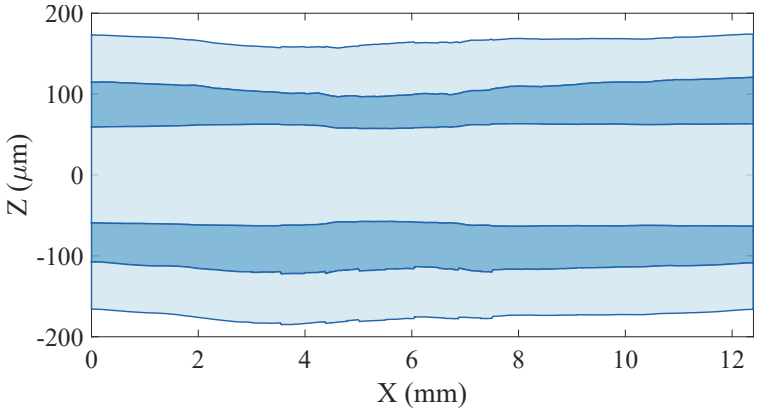

Fig. 17 Thickness profile for each layer in composite along length of web.

IV. Coiling of Longeron

A. Experimental Setup

It is of interest to validate the results of the numerical model in Section II.D by performing a coiling experiment on the thermoplastic longeron. Using similar equipment to a previous investigation [17], the composite longeron, trimmed to match the geometry of the numerical model, was secured to a coiling hub using an acrylic block and sandpaper, while tension was applied at the opposite end of the longeron with a weight and pulley system. In order to image and quantitatively measure the buckle that forms during coiling, DIC stereo-mounted cameras were used. The cameras and lights were mounted below the longeron aiming at the flange that would come into contact with the hub - as shown in Figure 18. This flange was prepared by first laying a thin matt-white layer of paint, followed by a speckle pattern using a black ink stamp with 0.66 mm wide black dots. Finally, the coiling hub was specifically designed for the test. It was fabricated with transparent acrylic tubing to reduce shading, and including a 90° cut-out to allow an unobstructed view of the buckle region towards the end of the test.

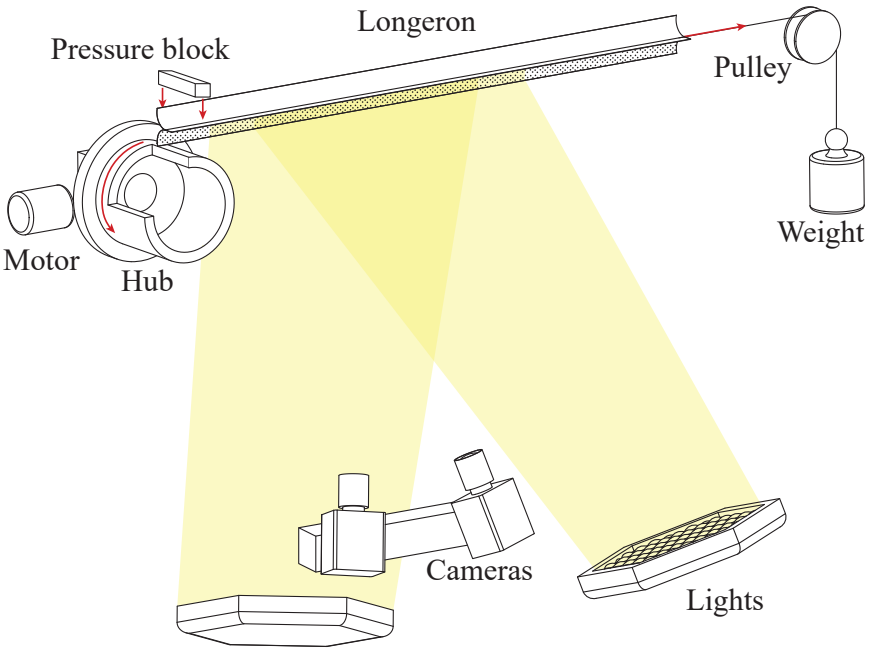

Fig. 18 Diagram of coiling test apparatus to measure buckle during coiling of longeron.

Images of the longeron during coiling were taken at a frequency of 2 Hz, which corresponds to every 0.5° of hub rotation, and were analyzed with the Correlated Solutions VIC 3D software to obtain the principal curvatures. The coiling experiment was stopped at a rotation of 265°, which was past the angle at which the buckle formed (identified as approximately 140° during the experiment) and allowed for the best view of the longeron through the cut-out in the acrylic tube. Raw images with the overlaid DIC data are presented in Figure 19. These same measured curvatures about the x and y-axis (defined in Section I as the transverse and longitudinal curvatures, respectively) in both the experiment and FEM are plotted in Figure 20 side-by-side for comparison, where the origin was set at the intersection between the edge of the flange (parallel to the y-axis) and the interface between the hub and the longeron (parallel to the x-axis). Thus, the left-hand side of the surface plots is the flange, while the right-hand side corresponds to the web. An evolution of the buckle is also presented in Figure 21, where the variation of K_x at both both points in the flange and the web is plotted. The flange location was selected to pass through the buckle and is thus located at the distance of 5 mm from the flange edge, while the web location was selected to be at 4 mm from the edge of the web.

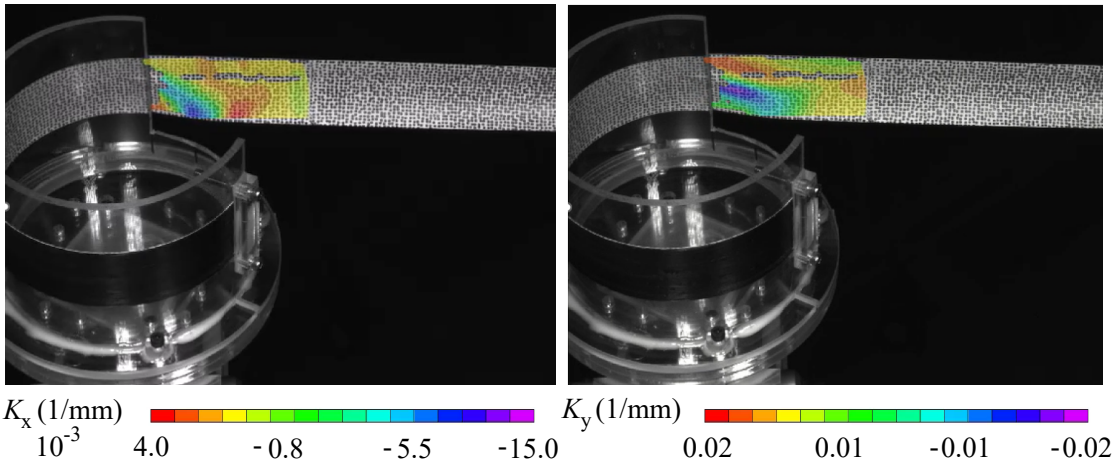

Fig. 19 Representative images of the measured curvatures around the hub-longeron transition region.

B. Discussion of Coiling Experiment Results

The surface plots of the transverse and longitudinal curvatures in Figure 20 show a close resemblance between the experiments and the finite element model and they clearly indicate that the DIC apparatus with the selected speckle size was able to measure some of the considerable curvature gradients that take place within the flange. The results are particularly similar when considering the flange, where both K_x plots depict the elongated shape of the buckled region corresponding to the negative curvature, which originates from the web-flange interface and traverses to the flange's edge at an approximate 40 ° angle. Both images show the increment in K_x that occurs above and below the buckle, even though the experiment presents a larger region with negative curvature, which spreads to the hub-longeron interface. When observing the K_y plots, there are however some significant differences. In particular, the FEM presents a series of longitudinal transition regions when moving across the cross-section in correspondence of the buckle. K_y is shown to vary from -0.03 mm^{-1} to 0.05 mm^{-1} to then -0.02 mm^{-1} in only about 20 mm, while the experiment has a single transition from -0.03 mm^{-1} to 0.02 mm^{-1} at the web-flange interface. Discrepancies in the transverse curvature were expected to a degree due to the shape imperfections measured in Section III.C. It was indeed shown that the imperfections were mostly along the length of the longeron, rather than localized in a specific region, and would therefore mostly affect the measured K_y. Considering the web-flange interface, there is more evidence for discrepancies in the μ-CT scans from Section III.D, which despite the seemingly sharp transition from six to three plies, show an approximately 20 μm gap between the laminates of the two flange. Instead, the FEM assumes that the flanges bend to form the web, with no gap at the interface.

Focusing on the evolution of K_x near the buckle as a function of the hub rotation, we can note similarities with the surface plots analyzed above. First, considering the flange, the curve indicates a more negative curvature near the center

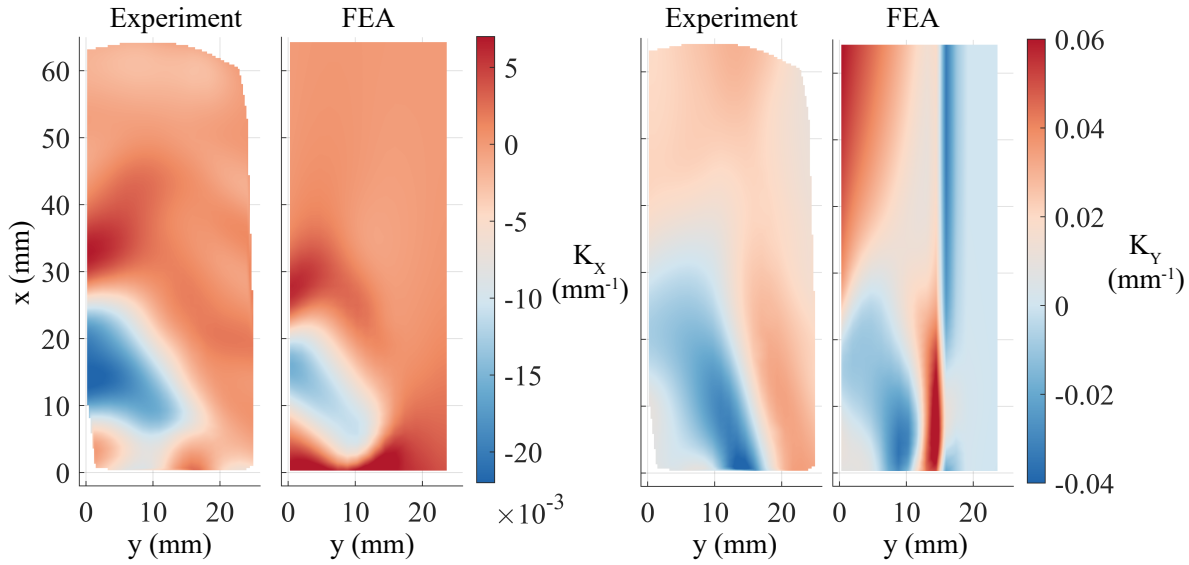

Fig. 20 K_x and K_y near buckled region in the bottom flange for both experiment and FEM. The y-axis is the flange edge and the x-axis is where the hub makes first contact with the longeron.

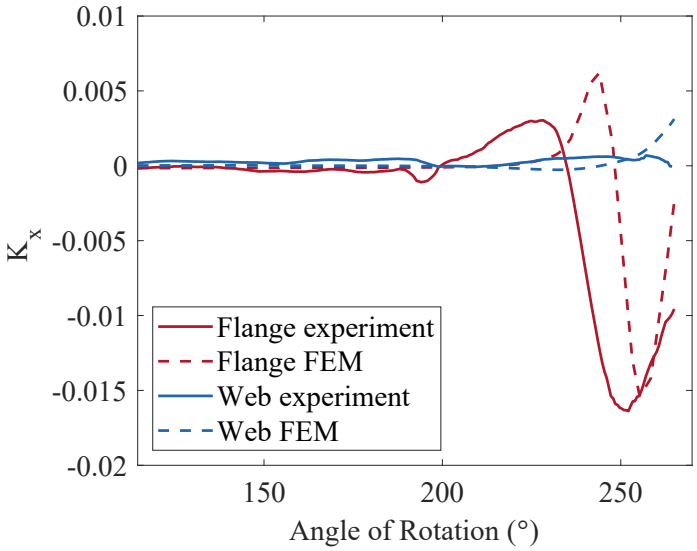

Fig. 21 Evolution of transverse curvature as a function of hub rotation.

of the buckle and a wider wavelength (defined as the width of the peak and trough) compared to the finite element model. This matches the larger width and height of the buckled region in the experiment-based surface plot compared to the numerical model. Second, focusing on the web, we can notice the curvature remains unchanged throughout the test while the finite element simulation shows an increment in curvature towards the end of the test. Because no buckle forms in the web, this indicates that, according to the numerical model, the web is starting to conform to the hub's radius before it makes contact with the hub itself as a result of an extended transition region before the hub. Instead, the experiment shows that the web conforms to the hub's radius only much closer to the contact interface. Both aspects of the comparison may be attributed to a laminate that is stiffer in the experiment than in the simulation. This discrepancy may be traced back to the choice of the ±45° tension test used to determine the material's shear stiffness, which does to not apply a pure shear load on the specimen. Similarly, the $[0/90/0]_s$ laminate implemented in the longeron was not directly characterized to obtain the properties for the model, while the standard CLT formulation of the ABD matrix was used to derive the laminate's properties. These inaccuracies in the input material properties

may therefore have affected the size and depth of the buckled region, and will be analyzed in more detail in the Section IV.C.

Despite these discrepancies, the experiments have confirmed an increment in local curvature in the buckle region in both longitudinal and transverse directions. In regards to K_x, neither flange nor web should present curvature before being deformed around the hub, yet in this region the material reaches a radius of 50 mm. Considering K_y, the material should be flattened and reach a curvature change of 13.6 mm, yet at the buckle this is closer to 38.6 mm according to the experiments, indicating an almost 3 fold increment in the curvature experienced by the material during coiling.

C. Effect of material properties on buckle shape in numerical model

A previous study [23] showed the significant effect that the shear stiffness moduli G_{12} of a composite laminate has on the formation and propagation of the buckle near the hub transition region. Thus, to better comprehend the difference between the FEA and experiment presented in Section IV.B, a number of simulations were performed with the same coiling model, but varying shear stiffness moduli. The geometry of the TRAC cross-section was left unchanged, while for this analysis the lamina thickness and coiling radius were kept at 50 μm and 50 mm, respectively. The shear modulus was modified from 25% of its original value of 5.1 GPa to 250%, covering a permissible range of values for a thermoplastic-based composite. The results for the simulations is presented in Figure 22, where 2D contour plots of the transverse curvature at four representative shear moduli are shown to display the shape and magnitude of the buckle. Below these is also a plot of the maximum measured curvature in the longeron, which for all simulations coincides with location 2 from Section II.E.

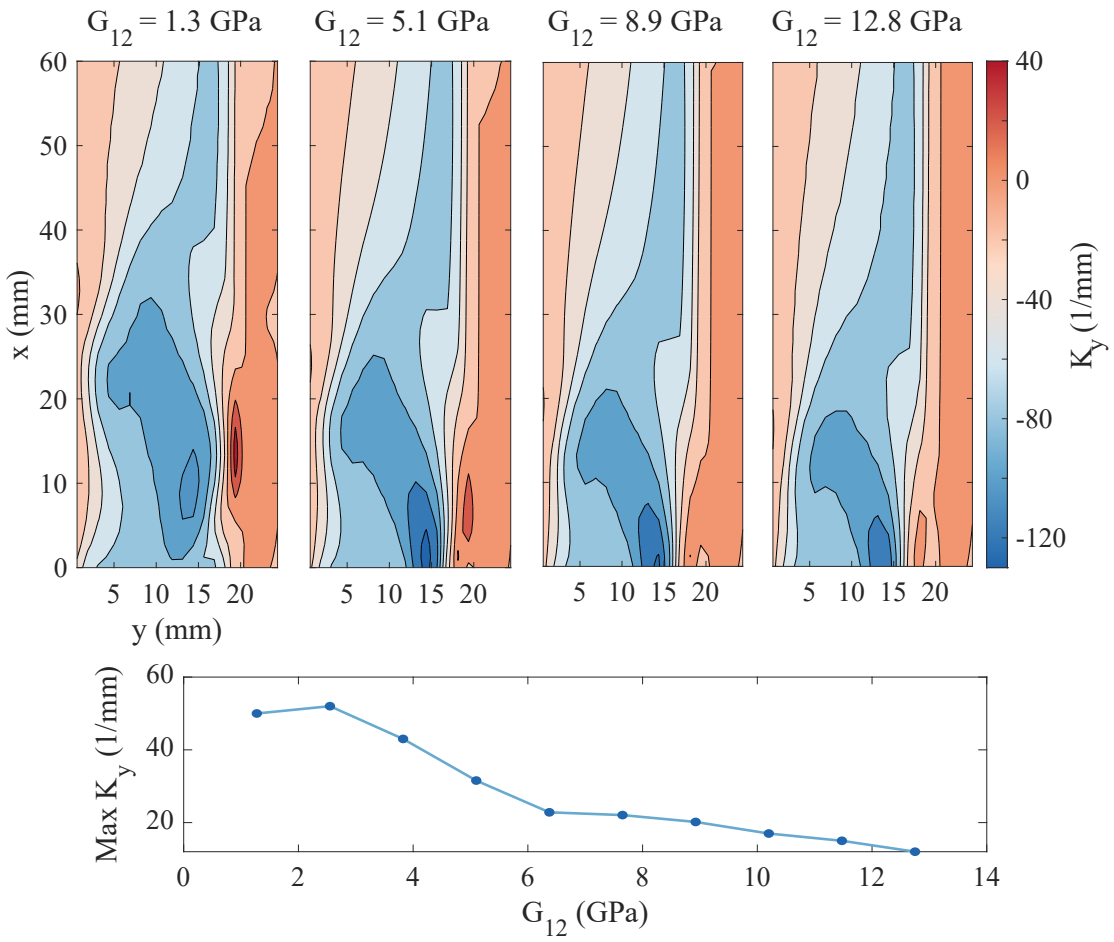

Fig. 22 Contours of K_y in the region affected by buckle formation for 4 different shear stiffness moduli and relationship between maximum measured K_y for varying shear stiffness moduli.

Comparing the contours, it is clear that an increase in stiffness reduces the size and location of the buckle. For a G_{12} of 1.3 GPa, the buckle extends for 30 mm in height and 15 mm in width and is located far from the longeron-hub transition because the softer material allows for a larger and more developed buckle to form. Instead, for a stiffness that is 10× larger the buckle is smaller, yet maintaining a similar shape, and its peak amplitude is closer to the transition region owing to the stiffer and thus less deformable material. Of greater greater interest, however, is the positive peak in curvature adjacent to the buckle and located at a y-value of 20 mm for all the simulations. This is a deformation in the material that is opposite to the buckle and is a result of the mismatch in coiling radius between the top and bottom flanges when coiled. Due to the bonding of the two flanges into the web, this mismatch in deformation generates the buckle and, as it is shown in these contours, if the shear moduli is low enough, the material is able to deform in the small space between the buckle and the web edge and exhibit this high positive K_y value. Instead, with increasing stiffness, this secondary deformation is limited and reduces to being negligible, as also observed in the experimental results.

This reduction is clearly shown the graph presenting the maximum curvature against shear stiffness, where a sharp drop occurs between 1.3 and 6.4 GPa, after which the measured curvature is still decreasing, but more slowly. These results indicate that the tested composite longeron has a large shear stiffness that what was measured through mechanical testing, possibly due to the suboptimal experimental method used to evaluate this property. In addition, it is clear that the effect of material stiffness on the buckle shape and induced concentrated stresses is not a straightforward relationship. Increasing stiffness causes a smaller deformation compared to a more compliant material. However, greater stiffness means that induced stresses for an equal bending curvature are greater and would trigger failure earlier. Thus, proper analysis of the coiling of such deployable structures is essential to prevent premature failure and undesirable plastic deformations.

V. Conclusion and Future Work

An ultra-thin thermoplastic-based prepreg has been characterized and used to design a cross-ply laminate that would not fail when built into coilable composite longerons. Analytical and numerical models were used to inform the design of the laminate, namely determining the relationship between laminate thickness and minimum coiling radius to avoid failure. A comparison of the results from these models clearly showed that the formation of buckles during coiling causes considerable underestimation of stresses by the analytical model and that the stresses due to this buckle vary non-linearly with respect to the coiling radius and lamina thickness. These results were the basis for designing and manufacturing a TPC TRAC longeron with what is, at the moment of writing, the thinnest available prepreg with PEEK matrix and spread-tow carbon fiber reinforcement.

This material was processed with a new two-stage manufacturing procedure, where stage 1 takes place in an autoclave - for initial compaction of the cross-ply laminas - and stage 2 in an oven - for shaping the laminas into the TRAC cross-section. The thermal cycles were selected to decrease the initial crystallinity of the material, thus reducing the temperature at which the composite material could be processed for reshaping. The resulting laminate was then evaluated at the macroscale, with laser scanning metrology equipment, and at the microscale, with μ-CT scanning. The former showed that the maximum deviation of the obtained longeron from nominal was 7% due to spring-back of the composite material from the shape of the mold, and that imperfections ran along the length of the longeron, indicating possible misalignment of the laminas or laminate during manufacturing. Instead, the CT scans showed a continuous interface between the flanges and the web, with generally uniform lamina thicknesses between all six plies and a total laminate thickness within the expected value based on UD laminates produced. Spectroscopy analysis of the manufactured longeron at different stages during processing also showed that the DoC was indeed kept lower, thus increasing the fraction of amorphous phase in the polymer that could be more easily reshaped into the TRAC cross-section.

Finally, a coiling experiment has demonstrated the validity of the numerical simulation, confirming the large gradients in curvatures taking place during the packaging of the longeron. In addition, small differences between the numerical model and experiment were attributed to the measured shear stiffness of the material, which greatly influences the curvature distribution in the hub-longeron transition region. The buckle was shown to be the limiting factor in the design of the laminate, and thus further work in mitigating its effects, through drop-ply laminates or a modified layup sequence, may be necessary. Similarly, further work may enable out-of-autoclave manufacturing of TPCs for thin shell structures applications. The current investigation has proved the feasibility of reshaping in an oven, yet the compaction

of the laminas in a continuous manner to rapidly manufacture large-scale components is a future objective. In addition, more robust evidence is needed to support the claim that the increased thickness of the middle layers in the web improves the fracture toughness under mode-I delamination, which may be obtained through appropriate mechanical testing.

Acknowledgments

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